



Blister Blaster

Home-Brewed Accessibility and Augmentation

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Short Exhibit Label / Overview

This exhibit explains how the Blister Blaster was created and the sociotechnical implications behind the creation. The Blister Blaster is an accessibility tool designed to help a person open medicinal blister packs. It allows people with arm or hand disabilities access medicine with less force and more control. This exhibit also examines the use of AI tools and 3D printing as enablers for customized accessibility augmentations.

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Background

The concept for the Blister Blaster began with my own personal struggles with blister medicine packs. I have problems with my nervous system that sometimes make fine motor movements impossible, especially with my hands. I found blister packs to be a pain even before I developed nerve issues. My old technique of opening blister packs involved some precision cutting with a knife, but that is a very bad idea now.

Existing accessibility aids for most medicine containers already exist. Reversible medicine lids (*Figure 1*), rubber gripping devices, medication alarms, and even “talking” labels exist to ensure that people have easy access to their medication. However, no commonly available device exists to augment access to blister medication packs. It is possible to find a few products, mostly in the form of 3D printer blueprints, but these products don’t address all my exact needs.

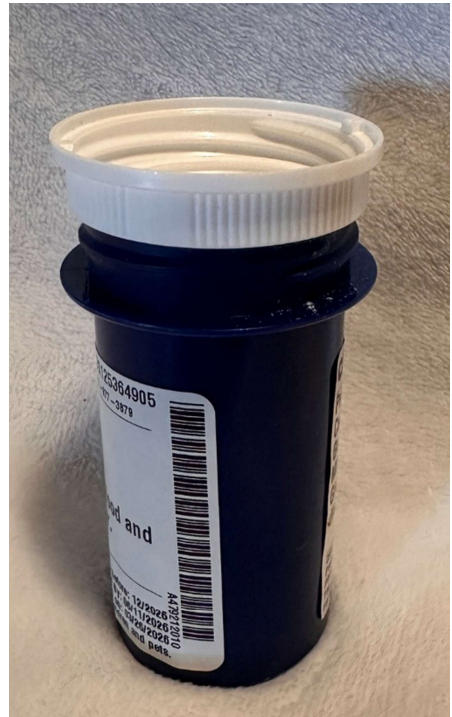


Figure 1: Example of an existing accessibility aid for a pill bottle. The lid can be inverted to give a larger gripping area.

What It Augments

The Blister Blaster augments a person’s hand strength, stability, and precision. The lever design enhances hand strength through mechanical advantage. Stability issues are resolved by making a simple design that requires minimal accuracy. The device also augments a user’s ability to successfully extract whole pills from the packaging without pulverizing them. Ultimately, the Blister Blaster allows the person to use it to become more independent.

Who It Is For

The Blister Blaster is intended for people who suffer from arthritis, chronic pain, nerve issues, or limited hand mobility. People with these conditions may find it difficult to access medicine in blister packs when they need it.

Individuals without any disabilities may also find the Blister Blaster to be a blessing. Caretakers such as doctors, nurses, and veterinarians who may be struggling with blister packs on a regular basis may find the Blister Blaster to be helpful. Finally, this tool may be useful to anyone who finds dealing with blister packs to be a nuisance.



Figure 2: Example of a typical blister pack.

How It Works

The Blister Blaster uses a lever positioned directly over the pill cavity as seen in *Figure 3*. When pressed, the lever focuses force onto a rounded plunger that pushes the pill cleanly through the foil. The tool helps to prevent crushing, stabilize the blister pack, and keeps fingers safely away from sharp edges or sudden pops.

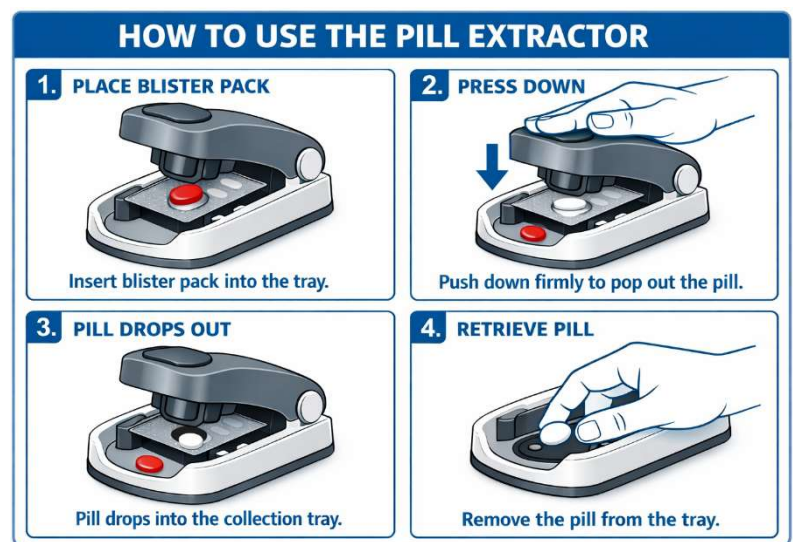


Figure 3: Instructions on how to use the Blister Blaster. This image was generated using Microsoft CoPilot.

Process & Critical Making

The first step of the process was to sketch my ideas. My first thought was to create a glove that had a thumb-mounted cutting surface to slice the blister pack open, and a small cup located in the palm to catch the pill (*Figure 4*). This turned out to be impractical due to the glove construction materials. However, this idea helped me focus on the needs, limitations, and abilities of the user, as well as the limitations of the project itself.

I settled on a press-down lever driven design (*Figure 5*) that seemed to be the most practical. The next step was to create proof of concept. I created a model out of a block of wood, a wood screw, and a hinge made from nail. This prototype worked surprisingly well but proved impractical for obvious safety reasons.

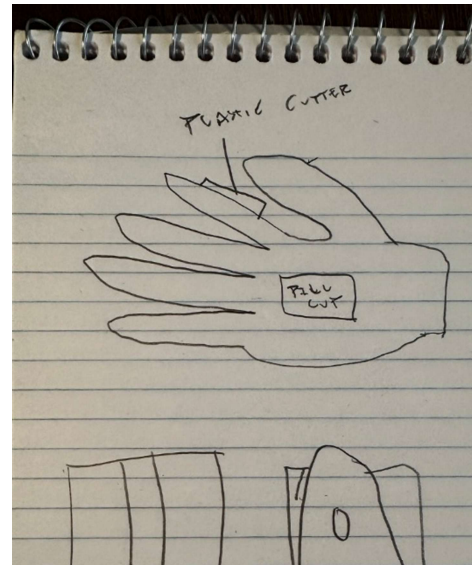


Figure 4: Initial sketch of a glove modified with a plastic cutter and small container designed to extract the medicine in blister

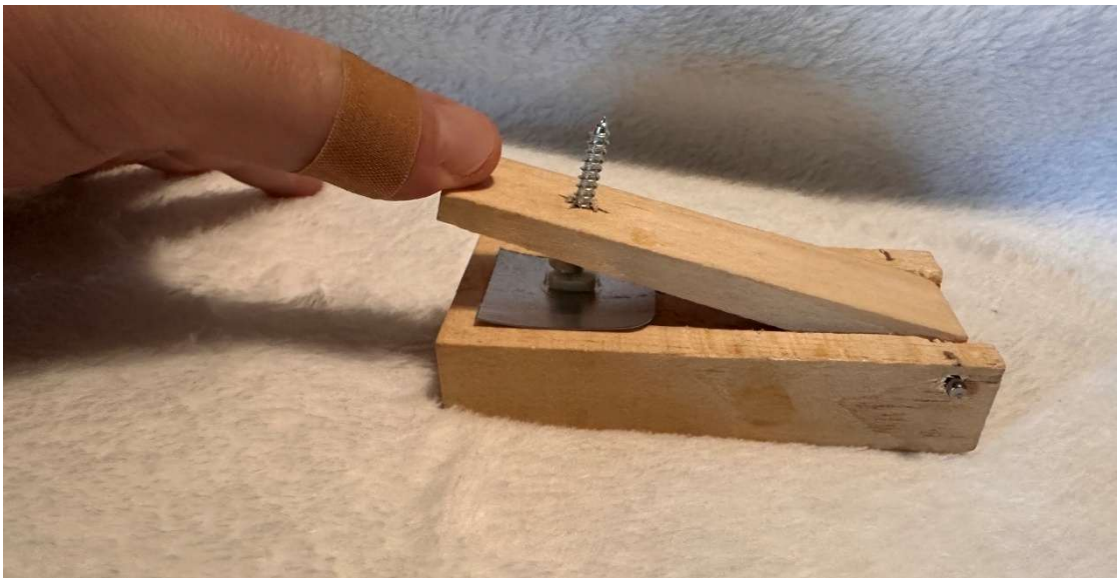


Figure 5: First physical prototype built. The body is made from balsa wood with a picture nail serving as the hinge and a screw used as a plunger.

From here I asked Microsoft CoPilot to create a 3D printable version of the Blister Blaster. Text prompting proved to be more successful than asking CoPilot to analyze my wooden prototype. However, the designs that CoPilot created (*Figure 6*) were lacking in several key areas. The long hinge would not be able to withstand the pressure of some blister packs, the hinge portions were too complicated, and the dimensions were too large.

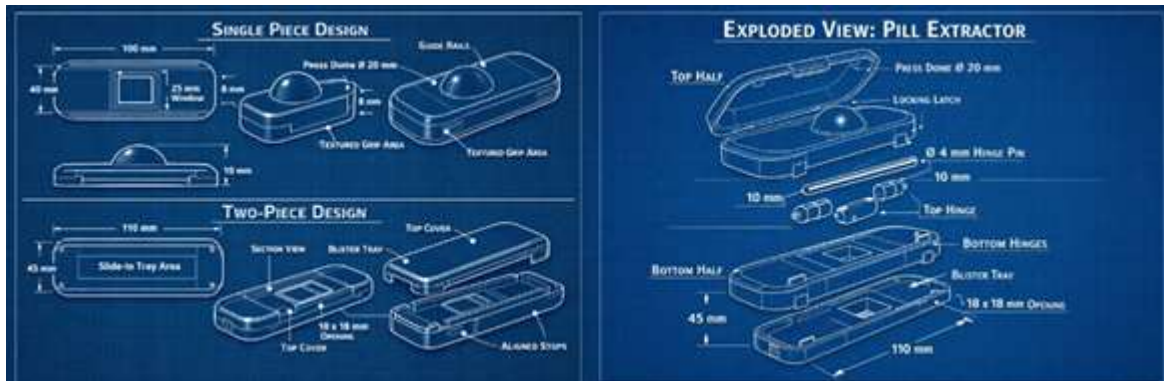


Figure 6: Microsoft Copilot designs after being prompted to "Create a device to extract pills from a blister pack for a user with arthritis and shaky hands. Create it with a lever/press mechanism and make it pocket-sized."

AI created several suggestions which I rejected for different reasons. I wanted to keep the complexity down, so anything over three assemblies was automatically rejected. I also wanted to make sure that it could be created by someone with limited dexterity, so I needed the parts to “snap” together. The designs that CoPilot created using only text proved to be impractical or impossible. After a few tries, I decided to upload a picture of my wooden prototype (Figure 5) to help the AI “see” what I was trying to create. With this new information CoPilot created a close approximation of what I wanted as seen in Figure 7.



Figure 7: Result of prompt to MS CoPilot that included a picture of the wooden prototype as an example.

From here I did some more refining to widen the handle and base and create a more accessible tray for the pill to fall into. After selecting the design that would ultimately become the Blister Blaster (Figure 8), I asked AI to create a 3D print file. Unfortunately, CoPilot struggled greatly with this task. The AI was able to generate 3D print files like the one in Figure 9, but the files did not mirror the design that I had picked.



Figure 8: The final design selected before moving into the 3D printing phase.

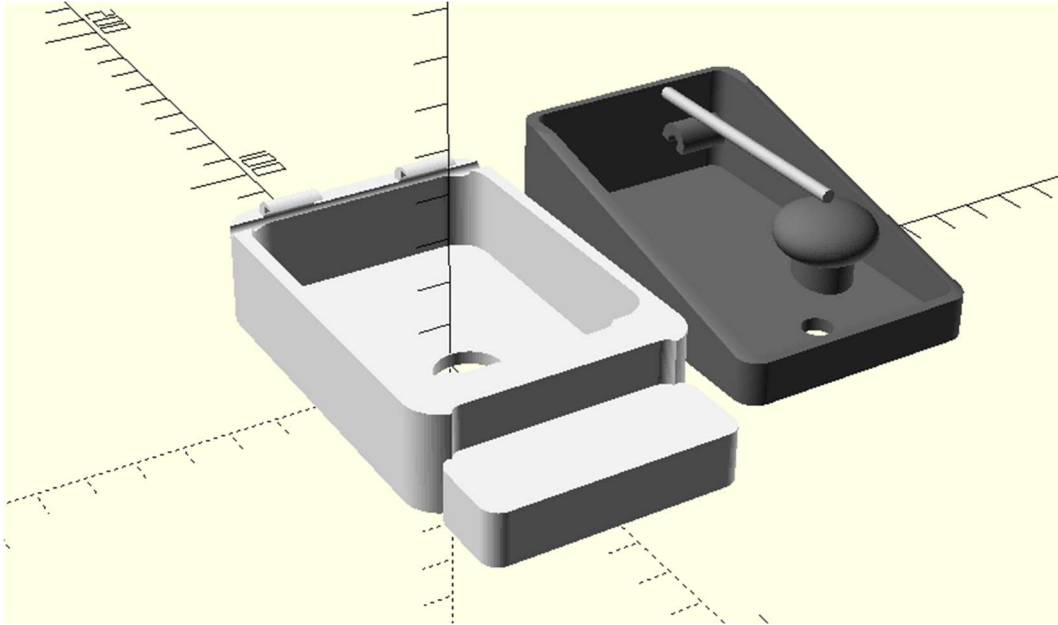


Figure 9 Example of 3D print file created using MS CoPilot. The AI was confident it could create 3D printer files using the design we created together, but it has fallen short.

Accessibility, Usability, & User Experience

Tests with the Blister Blaster were simple. Functionality was key. The tool should be able to function when used by individuals with hand and arm disabilities. Grip strength, shakiness, and chronic pain were the primary considerations when testing for accessibility and usability. Blister Blaster also needed to be able to extract medicine without pulverizing or crushing it due to pressure.

Additional considerations were made for ease of 3D printing and assembly of the device. This was done so that the disabilities addressed by the Blister Blaster did not hamper its creation and assembly. Designs with more than three pieces were discarded, as were any designs that contained smaller pieces. This proved to be problematic for the hinge, but the design was changed to a “snap” into place installation.

Testing feedback led to a few changes in the design of the Blister Blaster. The lever was made wider to allow the person to press down on it with a forearm or elbow. The removable tray was changed to an indent along the bottom plate. The plunger that pushes the medicine down and out was rounded to lessen the possibility of the medicine being crushed.

This project showed me that AI is incredibly helpful for designing an idea that may not be concrete enough. AI can fill in the gaps that one may be struggling with, especially during the brainstorming phase of creation. AI fell short in several areas, most of which came after the initial design. The AI was unable to create 3D print files that resembled and functioned the same way as the prototype it created. The AI also confused me by offering me too many options (*Figure 10*). If I had more technical knowledge or experience these options would have probably been helpful, but they ended up making a confusing situation even more difficult.

Option	What you get	What you need
OpenSCAD script	A <code>.scad</code> file with proper curves, hinge geometry, and the stapler shape	Free OpenSCAD app to render & export STL
Thingiverse/Printables search	Existing, tested, print-ready STL files that look like the render	Just download & slice
Technical drawing	Precise dimensions you can model yourself	Fusion 360, TinkerCAD, etc.

Figure 10: Example of options provided by Microsoft CoPilot for refining the 3D printer model.

3D printing allows users the ability to truly customize their designs for their specific needs. My wooden prototype was surprisingly effective despite its rudimentary build, but I wanted to make a few changes to make it more accessible. This would be difficult for me to create myself due to the pain, weakness, and shakiness in my hands. However, 3D printing and AI design are two tools that would ideally make these customizations a breeze.

Why It Matters

Access to medication should not be dependent on hand strength or mobility. Blister packs are a global standard, yet they exclude people with disabilities. Restoring a small part of a person's autonomy by being able to perform a simple task like opening their own medicine can feel like a huge accomplishment.

Augmentations such as these expand accessibility by removing usability barriers. In this case the barriers are physical disabilities, but in some cases the barriers aren't physical. It may be easier or more affordable for someone to access the internet and a 3D printer than it is to get items shipped to them. If a 3D printer is not available the blueprints will be, allowing a person to copy the design if nothing else. Perhaps they can even craft theirs out of wood.

The Blister Blaster and augmentations like it help shift the burdens of living from the body to the system. Instead of asking a disabled person to try and work around an intentionally poorly designed product, this allows that person to attack at least one common problem from a position of advantage. This advantage comes at a very cheap price. There are dozens, if not hundreds, of medical aids that are 3D printable free online.

The emergence of AI and 3D printing is something that allows someone to not only create their own medical aids from home but also tailor them to their exact disabilities or tastes. 3D printers are becoming more common inside the home and are available for use by the public at many public libraries. Printing is relatively cheap past the initial price of the printer itself. Even design has become easier with the assistance of AI. Artificial Intelligence can take an unformed idea and assist with the design and creation of the files needed to print it. This allows a user to easily design and print a truly personalized device.

Final Thoughts

The design and building of the Blister Blaster demonstrates how modern technology and tools expand accessibility and augmentation. Customization options are limited only by the capabilities of a 3D printer and the user's imagination. Artificial intelligence can assist the user in brainstorming, designing, and planning the creation of accessibility aids. These tools are not at the point where they are able to execute without significant user instruction, but they could easily reach the point of independent design in the next decade.

AI and 3D printing show us how augmentations can be democratized in the future. Allowing someone to design for their own bodies, limitations, and preferences is allowing that person to achieve independence. I realized that this project hints at a future where a person can create personalized accessibility devices quickly, cheaply, and at home. Accessibility in the future will stop being an afterthought, but rather it will be a process of personal exploration and empowerment.